

NORMAN WELLS AP, NT, Canada

WMO: 710430

Lat: 65.280N

Lon: 126.800W

Elev: 238

StdP: 14.57

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.0	-39.3	-49.4	0.3	-43.2	-46.3	0.4	-38.9	26.0	3.0	23.1	-0.3	1.4	90	0.748

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	81.7	62.7	78.2	61.2	74.8	59.7	64.5	78.0	62.7	74.6	61.1	72.0	8.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.5	76.5	68.2	57.8	71.9	67.0	56.0	67.3	66.0	29.7	78.2	28.4	74.7	27.3	72.2	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.6	18.8	17.1	-47.2	87.7	3.0	2.5	-49.4	89.5	-51.2	91.0	-52.9	92.4	-55.0	94.3
			-47.0	67.0	2.7	1.8	-49.0	68.4	-50.6	69.4	-52.1	70.5	-54.1	71.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.9	-13.7	-8.5	-0.9	22.6	44.5	59.1	62.7	56.7	44.3	25.7	1.3	-9.7	(6)
(7)		DBStd	30.11	15.42	13.61	13.54	12.70	10.21	7.34	6.26	7.35	7.75	8.83	12.21	13.60	(7)
(8)		HDD50	10546	1976	1637	1579	823	229	11	2	23	202	753	1461	1850	(8)
(9)		HDD65	15107	2441	2057	2044	1273	636	203	117	272	620	1218	1911	2315	(9)
(10)		CDD50	1002	0	0	0	0	59	285	395	231	32	0	0	0	(10)
(11)		CDD65	87	0	0	0	0	1	26	45	15	0	0	0	0	(11)
(12)		CDH74	872	0	0	0	0	31	279	408	152	2	0	0	0	(12)
(13)		CDH80	164	0	0	0	0	3	56	81	24	0	0	0	0	(13)
(14)	Wind	WSAvg	6.4	5.3	5.7	6.6	7.4	7.7	7.4	6.8	6.8	6.7	6.2	5.5	5.2	(14)
(15)	Precipitation	PrecAvg	13.1	0.8	0.7	0.5	0.6	0.8	1.6	2.2	2.2	1.2	1.2	0.7	0.8	(15)
(16)		PrecMax	20.2	2.4	2.8	1.5	2.5	2.6	4.1	4.6	4.6	3.4	3.4	1.5	2.0	(16)
(17)		PrecMin	7.3	0.0	0.0	0.0	0.0	0.0	0.3	0.5	0.1	0.2	0.2	0.1	0.2	(17)
(18)		PrecStd	3.0	0.5	0.6	0.4	0.5	0.5	0.8	1.0	1.2	0.6	0.7	0.4	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.2	28.5	40.9	59.7	77.9	85.8	86.7	83.5	71.0	50.4	33.7	27.3	(19)
MCWB			25.0	25.7	34.4	46.0	57.2	63.0	65.4	64.7	56.4	42.8	31.2	25.0	(20)	
2%		DB	19.1	20.0	33.6	51.8	72.2	80.9	82.0	78.5	65.3	43.5	26.6	19.0	(21)	
		MCWB	17.7	18.1	29.2	41.8	54.5	61.2	63.4	62.7	53.7	39.5	25.0	17.7	(22)	
5%		DB	12.6	14.5	25.3	46.1	67.3	77.0	78.6	74.0	61.1	39.9	21.5	13.8	(23)	
		MCWB	11.5	13.0	22.0	38.1	52.0	59.2	62.1	60.5	51.4	36.8	20.2	12.6	(24)	
10%		DB	6.8	10.0	19.3	41.6	62.2	73.3	75.2	69.8	57.1	36.9	17.9	8.5	(25)	
		MCWB	5.8	8.7	16.8	35.1	49.2	57.7	61.0	58.4	49.8	34.6	16.7	7.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.4	26.2	35.0	46.5	57.9	64.9	67.1	66.6	57.9	44.6	31.5	25.1	(27)
MDCB			28.3	28.3	40.4	58.9	76.3	80.8	81.5	79.7	67.6	48.4	33.5	27.1	(28)	
2%		WB	17.9	18.1	29.2	42.1	55.0	62.6	65.0	63.8	54.6	39.8	25.1	17.8	(29)	
		MDCB	19.1	19.8	33.9	51.1	70.8	77.6	78.3	76.4	63.0	43.1	26.5	19.0	(30)	
5%		WB	11.7	13.0	22.1	38.3	52.4	60.5	63.4	61.3	52.3	37.0	20.5	12.7	(31)	
		MDCB	12.8	14.5	25.2	45.7	66.3	74.2	75.3	71.5	60.0	39.6	21.6	13.9	(32)	
10%		WB	5.8	8.7	16.8	35.1	49.6	58.6	61.9	59.4	50.2	34.6	16.7	7.5	(33)	
		MDCB	6.8	10.0	19.1	41.1	61.2	70.9	72.7	68.1	56.4	36.8	17.8	8.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.6	15.4	20.0	20.5	20.2	19.6	18.8	17.2	15.6	9.9	11.6	12.6	(35)
MCDBR			18.1	18.9	25.8	23.2	28.3	25.1	24.5	24.3	23.6	14.4	13.3	16.9	(36)	
MCWBR			17.2	17.6	21.5	15.5	15.0	11.5	11.1	11.8	13.8	10.7	12.5	16.1	(37)	
5% WB		MCDBR	18.0	19.0	25.0	22.2	26.9	22.6	21.9	21.6	21.4	13.7	13.7	17.2	(38)	
		MCWBR	17.1	17.8	20.9	14.8	14.5	10.9	10.5	10.9	12.9	10.4	12.9	16.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.157	0.234	0.258	0.319	0.352	0.363	0.386	0.361	0.303	0.256	0.167	0.159	(40)	
(41)		taud	1.948	2.218	2.251	2.186	2.344	2.389	2.361	2.435	2.545	2.409	1.954	1.951	(41)	
(42)		Ebn at Noon	124	217	264	268	270	269	258	254	250	209	108	35	(42)	
(43)		Edh at Noon	6	16	25	35	33	33	33	28	20	13	6	2	(43)	
(44)	All-Sky Solar Radiation	RadAvg	41	246	783	1456	1746	1846	1574	1169	710	270	66	10	(44)	
(45)		RadStd	4	18	42	55	127	141	98	106	60	23	5	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.99	N/A	N/A	N/A	N/A	+1.01	-331	N/A	N/A	N/A	(46)
(47)	Regional (4 neighbors)	+1.33	+2.70	+2.50	N/A	N/A	N/A	-425	-416	N/A	N/A	(47)

Nomenclature: See separate page